

System Trimming Report: cstr_cooling_dynamics

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System Information

Number of States	3
Number of Inputs	1
State Variables	CA, T, Tc
Input Variables	u
Operating Conditions	{ 'T': 350 }
Parameters	{ 'q': 1.0, 'V': 1.0, 'CA0': 1.0, 'T0': 350.0, 'Tc0': 300.0, 'k

Equilibrium (Trim Point)

State Equilibrium (x_e):

CA	0.500017
T	350.000000
Tc	348.000068

Input Equilibrium (u_e):

u	2.083260
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Linearized Model

A matrix (State Matrix)

-1.9999	-0.0357	0.0000
836.7631	-180.3196	209.2051
0.0000	50.0000	-52.0833

B matrix (Input Matrix)

0.0000
0.0000
-48.0001

C matrix (Output Matrix)

1.0000	0.0000	0.0000
0.0000	1.0000	0.0000
0.0000	0.0000	1.0000

D matrix (Feedthrough Matrix)

0.0000
0.0000
0.0000

Stability Analysis

Classification: unstable

Poles / Eigenvalues:

Pole	Value	$ \lambda $
Pole 1	-236.8161	236.8161
Pole 2	-0.7703	0.7703
Pole 3	3.1835	3.1835

Diagnostics

Converged	Yes
Feasible	Yes
Timestamp	2026-02-12T15:23:39.792813